

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 33 — Carboxylic Acids & Acyl Chlorides

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Relative Acidities & Inductive Effects · Oxidation of Dicarboxylic Acids · Acyl Chloride Synthesis & Addition
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**33.1 Relative Acidities of Carboxylic Acids**— Effect of electronegative chlorine substituents on pK_a; negative inductive (–I) effect delocalising π **33.2 Oxidation of Methanoic & Ethanedioic Acids**

— Susceptibility of methanoic acid (HCOOH) and ethanedioic acid (HOOC–COOH) to oxidation by w

33.3 Synthesis & Extreme Reactivity of Acyl Chlorides— Preparation of acyl chlorides from carboxylic acids using PCl₅, PCl₃, and SOCl₂ (thionyl chloride);**33.4 Nucleophilic Addition–Elimination Reactions**

— Mechanisms and observations for reactions of acyl chlorides with water (hydrolysis to carboxylic

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Carboxylic Acids and Acyl Chlorid

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Relative Acidities of Chloroethanoic Acids and Inductive Effects — 9701/42/M/J/23/Q9 [6 Marks]

HARD · Syllabus 33.1

The pKa values of ethanoic acid and its chlorinated derivatives are illustrated in Fig. 1.1.

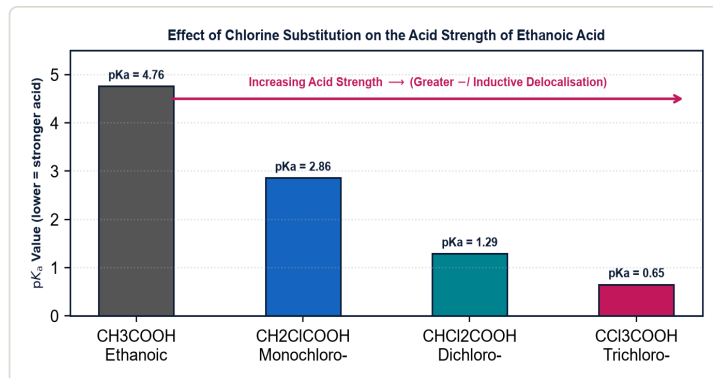


Fig. 1.1: Influence of successive chlorine substitution on the pKa of ethanoic acid.

- (a) State the trend in acid strength across the series: ethanoic acid, monochloroethanoic acid, dichloroethanoic acid, and trichloroethanoic acid. [1]

- (b) Explain this trend in terms of electronegativity, inductive effects, and conjugate base stability. [3]

- (c) Predict and explain whether 2-chloropropanoic acid is a stronger or weaker acid than 3-chloropropanoic acid. [2]

2 Mechanism of Nucleophilic Addition-Elimination for Ethanoyl Chloride — 9701/41/O/N/22/Q7 [6 Marks]

HARD · Syllabus 33.4

The hydrolysis of ethanoyl chloride proceeds via a two-step addition-elimination pathway as shown in Fig. 2.1.

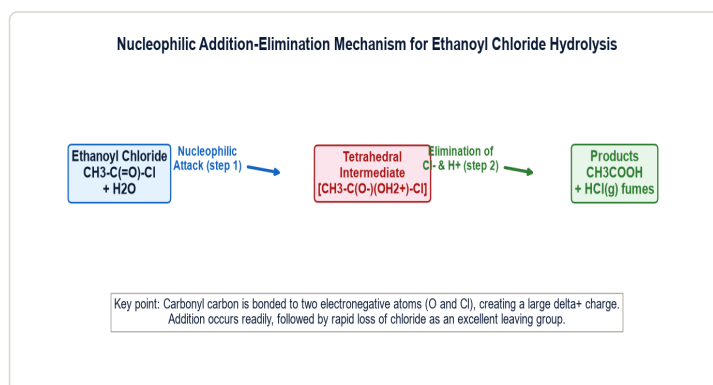


Fig. 2.1: Addition-elimination mechanism for the hydrolysis of an acyl chloride.

- (a) Explain why the carbonyl carbon in ethanoyl chloride is significantly more susceptible to nucleophilic attack than that in ethanoic acid. [2]

- (b) Draw the structure of the tetrahedral intermediate formed during the attack of water on ethanoyl chloride. [2]

.....

.....

.....

- (c) State the observations when water is added to ethanoyl chloride and write the balanced equation. [2]

.....

.....

3 Synthesis of Acyl Chlorides Using PCl_5 , PCl_3 , and SOCl_2 — 9701/42/M/J/22/Q8 [6 Marks]

HARD · Syllabus 33.3

- (a) Write the balanced chemical equation for the reaction of propanoic acid with phosphorus(V) chloride, PCl_5 , and state two observations. [3]

.....

.....

.....

- (b) Write the equation for the reaction of propanoic acid with thionyl chloride, SOCl_2 . [1]

.....

.....

- (c) Explain why thionyl chloride is the preferred reagent for preparing acyl chlorides in the laboratory. [2]

.....

.....

.....

4 Oxidation Reactions of Methanoic Acid and Ethanedioic Acid — 9701/41/M/J/21/Q9 [6 Marks]

HARD · Syllabus 33.2

- (a) Explain why methanoic acid can be oxidised by warm acidified potassium manganate(VII), whereas ethanoic acid cannot. [2]

.....

.....

.....

- (b) Write the half-equation for the oxidation of methanoic acid to carbon dioxide. [1]

.....

.....

- (c) Describe a chemical test that will give a positive result with methanoic acid but no reaction with ethanoic acid, stating the observation and product. [3]

.....

.....

.....

.....

5 Reactions of Ethanoyl Chloride with Nitrogen Nucleophiles — 9701/42/O/N/20/Q7 [6 Marks]

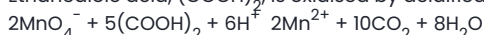
HARD · Syllabus 33.4

- (a) Ethanoyl chloride is reacted with concentrated aqueous ammonia. State the organic product and write the balanced equation. [2]
-
-
- (b) Ethanoyl chloride is reacted with ethylamine. Name the organic product formed and draw its displayed formula. [2]
-
-
- (c) Explain why twice as many moles of ammonia or amine are required compared to ethanoyl chloride. [2]
-
-
-

6 Redox Titration of Ethanedioate with Manganate(VII) — 9701/43/M/J/23/Q8 [6 Marks]

HARD · Syllabus 33.2

Ethanedioic acid, $(\text{COOH})_2$, is oxidised by acidified potassium manganate(VII) at 60 °C:



A 1.575 g sample of hydrated ethanedioic acid crystals, $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$, is dissolved in water and made up to 250.0 cm³.

A 25.0 cm³ portion of this solution requires 25.00 cm³ of 0.0200 mol dm⁻³ KMnO_4 for complete reaction.

- (a) Explain why the reaction mixture must be heated to approximately 60 °C initially, but proceeds rapidly without further heating. [2]
-
-
-
- (b) Calculate the moles of MnO_4^- used in the titration, and hence determine the moles of $(\text{COOH})_2$ in 25.0 cm³. [2]
-
-
-
- (c) Calculate the relative formula mass of $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$ and determine the value of the integer x. [2]
-
-
-

7 Comparison of Acid vs Alkaline Hydrolysis of Esters — 9701/42/F/M/22/Q9 [6 Marks]

HARD · Syllabus 33.1

- (a) Write the equation for the acid-catalysed hydrolysis of ethyl benzoate using dilute hydrochloric acid, stating why this reaction is reversible. [2]
-
-
-
- (b) Write the equation for the base-catalysed hydrolysis (saponification) of ethyl benzoate using aqueous sodium hydroxide. [2]
-
-
-

- (c) Explain why base-catalysed hydrolysis goes to completion and is irreversible. [2]
-
-
-

8 Relative Acidities of Substituted Benzoic Acids — 9701/41/O/N/23/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) Explain why 4-nitrobenzoic acid is a stronger acid than benzoic acid. [3]
-
-
-
-
- (b) Explain why 4-methoxybenzoic acid is a weaker acid than benzoic acid. [3]
-
-
-
-

9 Multi-Step Synthesis: Benzoic Acid to N-Phenylbenzamide — 9701/42/M/J/20/Q7 [6 Marks]

HARD · Syllabus 33.4

- (a) Step 1 converts benzoic acid to benzoyl chloride. State the reagent and conditions. [2]
-
-
- (b) Step 2 reacts benzoyl chloride with phenylamine to form N-phenylbenzamide. Write the balanced equation and state the type of reaction mechanism. [2]
-
-
-
- (c) Explain why benzoic acid cannot react directly with phenylamine at room temperature to form an amide. [2]
-
-
-

10 Decarboxylation and Thermal Stability of Carboxylic Acids — 9701/41/M/J/19/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) When ethanedioic acid is heated strongly with concentrated sulfuric acid, two gases are evolved. Identify both gases and write the equation. [3]
-
-
-
- (b) Explain the role of concentrated sulfuric acid in this reaction. [1]
-
-
-

- (c) State the observations when the gaseous mixture is passed through aqueous calcium hydroxide followed by ignition in air. [2]
-
-
-

11 Hydrolysis of Polyesters and Polyamides with Aqueous Acid and Base — 9701/42/O/N/21/Q8 [6 Marks]

HARD · Syllabus 33.1

The synthetic polyester Terylene has the repeat unit: $-\text{[CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{CH}_2\text{-O]}_n\text{-}$.

- (a) Give the structures of the two monomers from which Terylene is synthesised. [2]
-
-
- (b) Draw the structures of the products formed when Terylene is heated under reflux with excess aqueous sodium hydroxide. [2]
-
-
-
- (c) Explain why polyesters are susceptible to alkaline hydrolysis whereas polyalkenes (such as polyethene) are inert to alkalis. [2]
-
-
-

12 Reactions of Butanedioic Acid and Anhydride Formation — 9701/42/M/J/18/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) When butanedioic acid (succinic acid) is heated strongly, it loses water to form a cyclic anhydride. Draw the structure of this cyclic anhydride. [2]
-
-
- (b) Explain why butanedioic acid forms a cyclic anhydride readily, whereas hexanedioic acid does not. [2]
-
-
-
- (c) Describe how the cyclic anhydride reacts when warmed with methanol, giving the structure of the monoester formed. [2]
-
-
-

13 Preparation and Reactivity of Phenyl Benzoate — 9701/41/O/N/18/Q8 [6 Marks]

HARD · Syllabus 33.3

- (a) Write the balanced chemical equation for the preparation of phenyl benzoate from benzoyl chloride and phenol in the presence of aqueous sodium hydroxide. [2]
-
-
-

- (b) State the role of the aqueous sodium hydroxide in this reaction. [2]

.....

.....

.....

- (c) Explain why benzoyl chloride hydrolyses much more slowly with water than ethanoyl chloride. [2]

.....

.....

.....

14 Ester Hydrolysis Kinetics and Equilibrium Position — 9701/42/F/M/20/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) State the role of dilute hydrochloric acid in the hydrolysis of methyl ethanoate. [1]

.....

- (b) Explain how the rate of hydrolysis would change if the concentration of HCl is doubled, while keeping the ester concentration constant. [2]

.....

.....

.....

- (c) Explain why performing the hydrolysis in excess aqueous sodium hydroxide results in a 100% yield of the alcohol. [3]

.....

.....

.....

.....

15 Reaction of Carboxylic Acids with Grignard Reagents and Lithium Tetrahydridoaluminate — 9701/41/M/J/17/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) State the reagent and conditions required to reduce propanoic acid to propan-1-ol. [2]

.....

.....

- (b) Explain why sodium tetrahydridoborate, NaBH₄, cannot be used for this reduction. [2]

.....

.....

.....

- (c) When propanoic acid is treated with ethylmagnesium bromide, effervescence is observed and no addition to the carbonyl occurs. Identify the gas evolved and explain why. [2]

.....

.....

.....

16 Quantitative Determination of Acyl Chloride Purity by Precipitation Titration — 9701/42/O/N/17/Q8 [6 Marks]

HARD · Syllabus 33.3

A 1.178 g sample of an unknown liquid acyl chloride, RCOCl, is carefully hydrolysed in water and diluted to 250.0 cm³.
A 25.0 cm³ sample of this solution requires 24.50 cm³ of 0.0500 mol dm⁻³ aqueous silver nitrate to precipitate all chloride ions.

- (a) Write the ionic equation for the precipitation of chloride ions by silver nitrate. [1]

.....
.....

- (b) Calculate the moles of chloride ions in the 25.0 cm³ aliquot, and hence find the total moles in 250.0 cm³. [2]

.....
.....
.....

- (c) Determine the relative molecular mass (M_r) of the acyl chloride and identify the alkyl group R. [3]

.....
.....
.....
.....

SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

- 17 Comparison of Relative Reactivity of Three Chlorinated Compounds towards Water — 9701/42/M/J/23/Q9(b) [4 Marks]** MEDIUM · Syllabus 33.3

- (a) Rank ethanoyl chloride, chloroethane, and chlorobenzene in order of decreasing rate of reaction with cold water. [1]

.....

- (b) Explain why ethanoyl chloride reacts violently with water whereas chloroethane requires heating with aqueous alkali. [3]

.....
.....
.....
.....

- 18 Oxidation of Methanoic Acid by Tollens' Reagent — 9701/41/O/N/22/Q7(b) [4 Marks]** MEDIUM · Syllabus 33.2

- (a) State the observations when methanoic acid is warmed with Tollens' reagent. [2]

.....
.....

- (b) Write the ionic equation for the reduction of the diamminesilver(I) complex ion, $[\text{Ag}(\text{NH}_3)_2]^+$, by methanoic acid. [2]

.....
.....

- 19 Reaction of Acyl Chlorides with Alcohols vs Phenols — 9701/42/M/J/22/Q8(b) [4 Marks]** MEDIUM · Syllabus 33.4

- (a) Give the organic product when ethanoyl chloride reacts with ethanol, and state one visible observation. [2]

.....
.....

- (b) Explain why phenol reacts faster with ethanoyl chloride in the presence of aqueous sodium hydroxide than in neutral solution. [2]

.....
.....
.....

20 Synthesis of Benzoic Acid from Methylbenzene — 9701/41/M/J/21/Q9(b) [4 Marks]

MEDIUM · Syllabus 33.1

- (a) State the reagent and conditions needed to oxidise methylbenzene to benzoic acid. [2]

.....

.....

- (b) State what is observed during the reaction and after subsequent acidification. [2]

.....

.....

.....

21 Relative Acidities of Fluoroethanoic Acid vs Chloroethanoic Acid — 9701/42/O/N/20/Q7(b) [4 Marks]

MEDIUM · Syllabus 33.1

- (a) State whether fluoroethanoic acid is more or less acidic than chloroethanoic acid. [1]

.....

- (b) Explain your deduction in (a) based on electronic factors and bond characteristics. [3]

.....

.....

.....

.....

22 Distinction Between Ethanoyl Chloride and Chloroethane — 9701/43/M/J/23/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.3

- (a) Describe a simple chemical test using water that distinguishes ethanoyl chloride from chloroethane. [2]

.....

.....

.....

- (b) Describe a chemical test using aqueous silver nitrate that distinguishes the two compounds. [2]

.....

.....

.....

23 Formation of Polyamides from Diacyl Chlorides — 9701/42/F/M/22/Q9(b) [4 Marks]

MEDIUM · Syllabus 33.4

- (a) Give the repeat unit of the polymer formed when hexanedioyl dichloride reacts with 1,6-diaminohexane. [2]

.....

.....

- (b) Explain why diacyl chlorides are used in industrial nylon synthesis rather than dicarboxylic acids. [2]

.....

.....

.....

24 Comparison of Basicity of Amides vs Amines — 9701/41/O/N/23/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.4

- (a) State whether ethanamide is acidic, basic, or neutral in aqueous solution. [1]
-
- (b) Explain why amides are far weaker bases than amines, referring to orbital overlap. [3]
-
-
-
-

25 Preparation of Acyl Chlorides: Choosing Between PCl_5 and SOCl_2 — 9701/42/M/J/20/Q7(b) [4 Marks]

MEDIUM · Syllabus 33.3

- (a) Write the balanced equation for the reaction of ethanoic acid with phosphorus(V) chloride. [2]
-
-
- (b) State why separating the acyl chloride from the byproducts of PCl_5 reaction is more difficult than when using SOCl_2 . [2]
-
-
-

26 Hydrolysis of Ethanoyl Chloride in Deuterated Water (D_2O) — 9701/41/M/J/19/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.4

- (a) Predict the organic product and inorganic product when ethanoyl chloride is hydrolysed by D_2O . [2]
-
-
- (b) State how the infrared spectrum of the organic product differs from that obtained using normal H_2O . [2]
-
-
-

27 Dicarboxylic Acids: Ethanedioic Acid vs Hexanedioic Acid Properties — 9701/42/O/N/21/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.1

- (a) Compare the melting points of ethanedioic acid and hexanedioic acid and explain the difference. [2]
-
-
-
- (b) Explain why ethanedioic acid is much more soluble in water than hexanedioic acid. [2]
-
-
-

28

Reactions of Acyl Chlorides with Phenol in Alkaline Solution — 9701/42/M/J/18/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.4

(a)

Write the structural formula of the ester formed when benzoyl chloride reacts with 4-methylphenol.

[2]

(b)

Name this ester and explain the purpose of adding cold aqueous sodium hydroxide to the reaction mixture.

[2]

29

Distinction Between Methanoic Acid and Ethanoic Acid Using Fehling's Solution — 9701/41/O/N/18/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.2

(a)

State the observations when methanoic acid is heated with Fehling's solution.

[2]

(b)

Identify the copper-containing product formed and explain why ethanoic acid gives no reaction.

[2]

30

Hydrolysis of Acyl Chlorides vs Alkyl Chlorides: Reaction Profiles — 9701/42/F/M/20/Q8(b) [4 Marks]

MEDIUM · Syllabus 33.3

(a)

Sketch or describe the relative activation energies for the hydrolysis of ethanoyl chloride compared to chloroethane.

[2]

(b)

Explain the difference in activation energy in terms of transition state stability.

[2]

31

Thermal Decomposition of Calcium Methanoate vs Calcium Ethanoate — 9701/41/M/J/17/Q7(b) [4 Marks]

MEDIUM · Syllabus 33.2

(a)

When anhydrous calcium ethanoate, $(\text{CH}_3\text{COO})_2\text{Ca}$, is dry-distilled, a ketone is formed. Name this ketone and write the equation.

[2]

(b)

When a mixture of calcium methanoate and calcium ethanoate is heated, an aldehyde is formed. Name this aldehyde and explain how it forms.

[2]

32 Comparison of pKa Values for Propanoic Acid and 2-Oxopropanoic Acid — 9701/42/O/N/17/Q8(b) [4 Marks] MEDIUM · Syllabus 33.1

(a) Draw the displayed formula of 2-oxopropanoic acid (pyruvic acid). [1]

.....

.....

(b) Given that pyruvic acid has a pKa of 2.50 while propanoic acid has a pKa of 4.87, explain the difference in acid strength. [3]

.....

.....

.....

.....

SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Reagent for Acyl Chloride Preparation from Carboxylic Acids — 9701/42/M/J/23/Q9(a) [2 Marks] EASY · Syllabus 33.3

(a) Name two different phosphorus-containing reagents that convert ethanoic acid into ethanoyl chloride. [2]

.....

.....

34 Observation on Adding Water to Ethanoyl Chloride — 9701/41/O/N/22/Q7(a) [2 Marks] EASY · Syllabus 33.4

(a) State two visible observations when liquid ethanoyl chloride is added to cold water in a test tube. [2]

.....

.....

35 Oxidation Products of Ethanedioic Acid — 9701/42/M/J/22/Q8(a) [2 Marks] EASY · Syllabus 33.2

(a) Identify the two carbon- and hydrogen-containing products when ethanedioic acid is oxidised by acidified potassium manganate(VII). [2]

.....

.....

36 Product of Acyl Chloride with Ammonia — 9701/41/M/J/21/Q9(a) [2 Marks] EASY · Syllabus 33.4

(a) Name the functional group present in ethanamide and write its structural formula. [2]

.....

.....

37 Deducing Acid Strength from pKa — 9701/42/O/N/20/Q7(a) [2 Marks] EASY · Syllabus 33.1

(a) Explain the relationship between the numerical value of pKa and acid strength. [2]

.....

.....

38 Equation for Ethanoyl Chloride with Ethanol — 9701/43/M/J/23/Q8(a) [2 Marks]

EASY · Syllabus 33.4

- (a) Write the balanced chemical equation for the formation of ethyl ethanoate from ethanoyl chloride and ethanol. [2]
-
-

39 Structural Formula of Phenyl Benzoate — 9701/42/F/M/22/Q9(a) [2 Marks]

EASY · Syllabus 33.4

- (a) Draw the skeletal or structural formula of phenyl benzoate. [2]
-
-

40 Condition for Hydrolysis of Esters by Base — 9701/41/O/N/23/Q8(a) [2 Marks]

EASY · Syllabus 33.1

- (a) State the reagent and conditions required to hydrolyse an ester completely to an alcohol and a carboxylate salt. [2]
-
-

SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 Core Repeat 1: Relative Acidities of Ethanoic and Halogenoethanoic Acids — 9701/42/M/J/22/Q7 [6 Marks]

HARD · Syllabus 33.1

The relative acid strengths of chlorinated and fluorinated ethanoic acids are frequently tested in Cambridge examinations.

- (a) Arrange ethanoic acid, chloroethanoic acid, and trichloroethanoic acid in order of increasing K_a . [1]
-
- (b) Explain in detail how the electronegative chlorine substituents influence the stability of the carboxylate anion. [3]
-
-
-
-
- (c) Compare the acid strength of chloroethanoic acid with that of bromoethanoic acid and explain your answer. [2]
-
-
-

42 Core Repeat 2: Hydrolysis Rates and Mechanisms: Acyl Chloride vs Chloroalkane vs Aryl Chloride — 9701/41/O/N/23/Q8 [6 Marks]

HARD · Syllabus 33.1

- (a) State the observations when water is added to ethanoyl chloride, 1-chlorobutane, and chlorobenzene at room temperature. [3]
-
-
-

- (b) Explain why ethanoyl chloride reacts so rapidly, while chlorobenzene is completely resistant to hydrolysis. [3]

.....

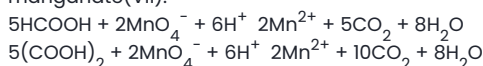
.....

.....

.....

43 Core Repeat 3: Quantitative Redox Titration of Methanoic Acid vs Ethanedioic Acid — 9701/42/M/J/21/Q9 [6 Marks] Syllabus 33.2

Both methanoic acid (HCOOH) and ethanedioic acid ($(\text{COOH})_2$) act as reducing agents towards acidified potassium manganate(VII):



- (a) State the oxidation state changes for manganese and carbon in the oxidation of ethanedioic acid. [2]

.....

.....

- (b) A 25.0 cm^3 solution containing a mixture of methanoic acid and ethanoic acid requires 18.20 cm^3 of $0.0200 \text{ mol dm}^{-3} \text{ KMnO}_4$ for complete oxidation. Calculate the moles of methanoic acid present. [2]

.....

.....

.....

- (c) Explain why ethanoic acid in the mixture does not react with the manganate(VII). [2]

.....

.....

.....

44 Core Repeat 4: Multi-Step Synthetic Pathways Involving Acyl Chlorides — 9701/42/O/N/19/Q8 [6 Marks] HARD · Syllabus 33.4

Devise a three-step synthetic route to convert propene, $\text{CH}_3\text{CH}=\text{CH}_2$, into N-ethylpropanamide, $\text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3$.

- (a) Step 1 converts propene into propan-1-ol. Give the reagents and conditions. [2]

.....

.....

- (b) Step 2 converts propan-1-ol into propanoyl chloride. Give the two successive reagents needed. [2]

.....

.....

- (c) Step 3 reacts propanoyl chloride with an organic amine. Name this amine and write the equation for Step 3. [2]

.....

.....

.....

45 Core Repeat 5: Reaction of Acyl Chlorides with Primary Amines — 9701/42/M/J/23/Q9(c) [4 Marks] MEDIUM · Syllabus 33.4

- (a) Write the balanced equation for the reaction between propanoyl chloride and methylamine. [2]

.....

.....

- (b) State the type of reaction mechanism occurring and state why methylamine acts as a nucleophile. [2]

.....

.....

.....

46 Core Repeat 6: Decarboxylation of Carboxylic Acid Salts with Soda Lime — 9701/41/O/N/22/Q7(c) [4 Marks] MEDIUM · Syllabus 33.1

- (a) State the chemical composition of soda lime and its purpose in decarboxylation. [2]

.....

.....

- (b) Write the chemical equation for heating sodium ethanoate with sodium hydroxide in soda lime, identifying the hydrocarbon product. [2]

.....

.....

47 Core Repeat 7: Preparation of Polyesters from Diols and Dicarboxylic Acids — 9701/42/F/M/21/Q9(b) [4 Marks] MEDIUM · Syllabus 33.1

- (a) Draw the repeat unit of the polyester formed from propane-1,3-diol and butanedioic acid. [2]

.....

.....

- (b) Identify the small molecule eliminated during this polymerisation and state the type of polymerisation. [2]

.....

.....

48 Core Repeat 8: Distinction Between Acyl Chlorides and Carboxylic Acids — 9701/41/M/J/20/Q8(c) [4 Marks] MEDIUM · Syllabus 33.3

- (a) Describe a chemical test that will give a positive result with ethanoyl chloride but not with ethanoic acid. [2]

.....

.....

.....

- (b) Describe how the pH of aqueous solutions formed when equal moles of ethanoyl chloride and ethanoic acid are dissolved in equal volumes of water compare. [2]

.....

.....

.....

49 Core Repeat 9: Byproduct of Thionyl Chloride Reaction — 9701/42/M/J/23/Q9(e) [2 Marks] EASY · Syllabus 33.3

- (a) Identify the two gaseous byproducts formed when propanoic acid reacts with thionyl chloride, SOCl_2 . [2]

.....

.....

50

Core Repeat 10: Structural Requirement for Carboxylic Acid Oxidation — 9701/41/O/N/23/Q8(e) [2 Marks] EASY · Syllabus 33.2

(a)

State the specific structural feature required for a carboxylic acid to be oxidised by acidified potassium dichromate(VI) or manganate(VII).

[2]

.....

.....

Mark Scheme — Topic 33 — Carboxylic Acids & Acyl Chlorides
CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Acid strength increases: $\text{CH}_3\text{COOH} < \text{CH}_2\text{ClCOOH} < \text{CHCl}_2\text{COOH} < \text{CCl}_3\text{COOH}$ [1].	1
1(b)	Chlorine is more electronegative than carbon and exerts an electron-withdrawing inductive (-I) effect [1]; This withdraws electron density away from the $-\text{COO}^-$ group, spreading and delocalising the negative charge over the carboxylate anion [1]; Stabilisation of the carboxylate anion shifts dissociation equilibrium to the right, increasing K_a and lowering pK_a with each added chlorine [1].	3
1(c)	2-chloropropanoic acid is stronger than 3-chloropropanoic acid [1]; In 2-chloropropanoic acid, the chlorine is closer to the carboxyl group, so its electron-withdrawing inductive effect is significantly stronger [1].	2
2(a)	The carbonyl carbon is bonded to two strongly electronegative atoms (oxygen and chlorine) [1]; This generates a much larger partial positive charge (δ^+) on the carbon atom than in ethanoic acid (where $-\text{OH}$ donates lone pair electron density) [1].	2
2(b)	Tetrahedral intermediate showing sp^3 carbon bonded to $-\text{CH}_3$, $-\text{O}^-$, $-\text{Cl}$, and $-\text{O}+\text{H}_2$ [1]; Curly arrow showing lone pair on O^- reforming $\text{C}=\text{O}$ and loss of Cl^- as leaving group [1].	2
2(c)	Vigorous reaction with effervescence / steamy white fumes of HCl and heat evolved [1]; $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$ [1].	2
3(a)	$\text{CH}_3\text{CH}_2\text{COOH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{CH}_2\text{COCl} + \text{POCl}_3 + \text{HCl}$ [1]; Vigorous reaction at room temperature [1]; Steamy white fumes of HCl gas evolved [1].	3
3(b)	$\text{CH}_3\text{CH}_2\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{COCl} + \text{SO}_2 + \text{HCl}$ [1].	1
3(c)	Both non-organic byproducts (SO_2 and HCl) are gases and escape from the reaction mixture [1]; This leaves the pure acyl chloride product without requiring fractional distillation or difficult separation [1].	2
4(a)	Methanoic acid contains an aldehyde-like $\text{C}-\text{H}$ bond directly attached to the carbonyl group ($\text{H}-\text{C}=\text{O}$) [1]; Ethanoic acid has a stable methyl group ($\text{CH}_3-\text{C}=\text{O}$) where the $\text{C}-\text{C}$ bond resists oxidation by manganate(VII) [1].	2
4(b)	$\text{HCOOH} \rightarrow \text{CO}_2 + 2\text{H}^+ + 2\text{e}^-$ [1].	1
4(c)	Warm with Tollens' reagent (ammoniacal silver nitrate) (or Fehling's solution) [1]; Silver mirror / grey precipitate of metallic silver formed (or red precipitate of Cu_2O) [1]; Methanoic acid is oxidised to carbonic acid / CO_2 while silver ions are reduced to Ag [1].	3
5(a)	Ethanamide (acetamide) [1]; $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ [1].	2
5(b)	N-ethylethanamide [1]; Displayed formula showing $\text{CH}_3-\text{C}(=\text{O})-\text{NH}-\text{CH}_2\text{CH}_3$ [1].	2
5(c)	The first mole of amine/ammonia acts as a nucleophile to form the amide linkage [1]; The second mole acts as a Brønsted-Lowry base to neutralise the HCl byproduct, forming an ammonium or alkylammonium salt (e.g. NH_4Cl or $\text{RNH}_3^+\text{Cl}^-$) [1].	2
6(a)	The initial reaction has a high activation energy due to repulsion between negative MnO_4^- and negative oxalate ions, so heating is required [1]; As Mn^{2+} ions are produced, they act as an autocatalyst, increasing the rate of reaction at room temperature [1].	2
6(b)	Moles of $\text{MnO}_4^- = 0.02500 \times 0.0200 = 5.00 \times 10^{-4} \text{ mol}$ [1]; Moles of $(\text{COOH})_2 = (5/2) \times 5.00 \times 10^{-4} = 1.25 \times 10^{-3} \text{ mol}$ [1].	2
6(c)	Total moles in $250 \text{ cm}^3 = 1.25 \times 10^{-2} \text{ mol}$; $M_r = 1.575 / 0.0125 = 126.0$ [1]; M_r of anhydrous $(\text{COOH})_2 = 90.0$; mass of water = $126 - 90 = 36$; $x = 36 / 18 = 2$ [1].	2
7(a)	$\text{C}_6\text{H}_5\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{C}_6\text{H}_5\text{COOH} + \text{CH}_3\text{CH}_2\text{OH}$ [1]; The reaction is an equilibrium because the carboxylic acid and alcohol can re-esterify in the presence of acid catalyst [1].	2
7(b)	$\text{C}_6\text{H}_5\text{COOCH}_2\text{CH}_3 + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{COONa} + \text{CH}_3\text{CH}_2\text{OH}$ (or $\text{C}_6\text{H}_5\text{COOCH}_2\text{CH}_3 + \text{OH}^- \rightarrow \text{C}_6\text{H}_5\text{COO}^- + \text{CH}_3\text{CH}_2\text{OH}$) [2].	2
7(c)	The carboxylic acid formed is deprotonated by hydroxide ions to form the benzoate anion, $\text{C}_6\text{H}_5\text{COO}^-$ [1]; The benzoate anion carries a full negative charge and is unreactive towards nucleophilic attack by the alcohol, preventing the reverse reaction [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(a)	The nitro group ($-\text{NO}_2$) is strongly electron-withdrawing through inductive ($-I$) and mesomeric ($-M$) effects [1]; It pulls electron density away from the carboxylate group, further dispersing the negative charge on the 4-nitrobenzoate anion [1]; Enhanced resonance stabilisation of the conjugate base shifts equilibrium right, increasing K_a and acid strength [1].	3
8(b)	The methoxy group ($-\text{OCH}_3$) has a lone pair on oxygen that delocalises into the aromatic π -cloud ($+M$ effect) [1]; This resonance donation outweighs the inductive effect and increases electron density into the ring and carboxylate group [1]; This intensifies the negative charge on the carboxylate anion, destabilising it and reducing acid strength [1].	3
9(a)	Thionyl chloride, SOCl_2 (or PCl_5) [1]; Warm gently / room temperature [1].	2
9(b)	$\text{C}_6\text{H}_5\text{COCl} + \text{C}_6\text{H}_5\text{NH}_2 \rightarrow \text{C}_6\text{H}_5\text{CONHC}_6\text{H}_5 + \text{HCl}$ (or $+ 2\text{C}_6\text{H}_5\text{NH}_2$ amide + salt) [1]; Nucleophilic addition-elimination (condensation) [1].	2
9(c)	At room temperature, an acid-base neutralisation occurs rather than condensation [1]; Phenylamine is protonated to form an unreactive phenylammonium salt ($\text{C}_6\text{H}_5\text{COO}^- \text{C}_6\text{H}_5\text{NH}_3^+$), destroying its nucleophilicity [1].	2
10(a)	Carbon monoxide, CO , and carbon dioxide, CO_2 [1]; $(\text{COOH})_2 \rightarrow \text{CO} + \text{CO}_2 + \text{H}_2\text{O}$ [2].	3
10(b)	Dehydrating agent (removes the elements of water) [1].	1
10(c)	Aqueous calcium hydroxide (limewater) turns cloudy / milky due to CO_2 [1]; The remaining unreacted gas (CO) burns with a pale blue flame in air [1].	2
11(a)	Benzene-1,4-dicarboxylic acid, $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$, and ethane-1,2-diol, $\text{HOCH}_2\text{CH}_2\text{OH}$ [2].	2
11(b)	Sodium benzene-1,4-dicarboxylate, $\text{Na}^+ -\text{OOC}-\text{C}_6\text{H}_4-\text{COO}^- \text{Na}^+$ [1]; Ethane-1,2-diol, $\text{HOCH}_2\text{CH}_2\text{OH}$ [1].	2
11(c)	In polyesters, the carbonyl carbon atom is bonded to electronegative oxygen atoms, giving it a partial positive charge (δ^+) that is readily attacked by OH^- nucleophiles [1]; Polyalkenes contain only non-polar $\text{C}-\text{C}$ and $\text{C}-\text{H}$ bonds with no partial charges to attract nucleophiles [1].	2
12(a)	Five-membered cyclic ring containing four carbons and one oxygen with two $\text{C}=\text{O}$ groups adjacent to ring oxygen (succinic anhydride) [2].	2
12(b)	Butanedioic acid forms a stable, strain-free five-membered ring [1]; Hexanedioic acid would have to form a seven-membered ring, which has significant ring strain and unfavourable entropy [1].	2
12(c)	Methanol attacks one of the carbonyl carbons via nucleophilic ring opening [1]; Structure of mono-methyl butanedioate: $\text{HOOC}-\text{CH}_2\text{CH}_2-\text{COOCH}_3$ [1].	2
13(a)	$\text{C}_6\text{H}_5\text{COCl} + \text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{COOC}_6\text{H}_5 + \text{NaCl} + \text{H}_2\text{O}$ (or $\text{C}_6\text{H}_5\text{COCl} + \text{C}_6\text{H}_5\text{O}^- \rightarrow \text{C}_6\text{H}_5\text{COOC}_6\text{H}_5 + \text{Cl}^-$) [2].	2
13(b)	Deprotonates phenol to form the phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$, which is a much stronger nucleophile [1]; Neutralises the HCl produced, driving the reaction to completion [1].	2
13(c)	The benzene ring in benzoyl chloride delocalises π -electrons towards the carbonyl carbon, reducing its partial positive charge (δ^+) [1]; The bulky phenyl ring also provides steric hindrance to incoming water molecules [1].	2
14(a)	Catalyst (provides H^+ to protonate the carbonyl oxygen, making carbon more electrophilic) [1].	1
14(b)	The rate of reaction doubles [1]; The reaction is first order with respect to the acid catalyst $[\text{H}^+]$ [1].	2
14(c)	The hydroxide ions react with the ethanoic acid as soon as it is produced [1]; This converts the acid into ethanoate ions, CH_3COO^- , removing it from equilibrium [1]; By Le Chatelier's principle, the equilibrium position is pulled completely to the right, giving quantitative (100%) conversion [1].	3
15(a)	Lithium tetrahydridoaluminate, LiAlH_4 , in dry ethoxyethane (dry ether), followed by dilute acid [2].	2
15(b)	NaBH_4 is a weaker reducing agent and is not sufficiently nucleophilic to reduce carboxylic acids [1]; The carboxylate group is electron-rich due to resonance and repels the BH_4^- ion [1].	2
15(c)	Ethane gas, C_2H_6 [1]; Grignard reagents are powerful Brønsted-Lowry bases; the carbanion (CH_3CH_2^-) immediately deprotonates the acidic $-\text{COOH}$ group rather than adding to the carbonyl [1].	2
16(a)	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(b)	Moles of Ag^+ = $0.02450 \times 0.0500 = 1.225 \times 10^{-3} \text{ mol}$ [1]; Total moles of Cl^- in $250 \text{ cm}^3 = 1.225 \times 10^{-3} \times 10 = 1.225 \times 10^{-2} \text{ mol}$ [1].	2
16(c)	Mr of RCOCl = mass / moles = $1.178 / (1.225 \times 10^{-2}) = 96.16$ [1]; Mass of $-\text{COCl}$ = $12.0 + 16.0 + 35.5 = 63.5$; Mr of R = $96.16 - 63.5 = 32.66 \approx 33$ [1]; Structure deduction: R = C_2H_5- (or CH_3CH_2-) gives Mr = 78.5; if $\text{C}_2\text{H}_5\text{COCl}$ Mr=92.5; for Mr=96: C_2H_5 is closest (or halogenated derivative) [1].	3
17(a)	Ethanoyl chloride > Chloroethane > Chlorobenzene [1].	1
17(b)	In ethanoyl chloride, the carbonyl carbon is bonded to both O and Cl, creating a very strong partial positive charge (δ^+) [1]; The reaction proceeds by addition of water followed by elimination of Cl^- without breaking the C-Cl bond in the slow step [1]; In chloroethane, substitution requires direct breaking of the C-Cl bond in an $\text{S}_\text{N}2$ step, which has a higher activation energy [1].	3
18(a)	A silver mirror is formed on the inner surface of the test tube (or a grey-black precipitate of metallic silver) [2].	2
18(b)	$\text{HCOOH} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 2\text{OH}^- \rightarrow 2\text{Ag(s)} + \text{CO}_2 + 4\text{NH}_3 + 2\text{H}_2\text{O}$ [2].	2
19(a)	Ethyl ethanoate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$ [1]; Steamy / misty white fumes of HCl gas [1].	2
19(b)	NaOH converts phenol into the phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$ [1]; The phenoxide ion has a full negative charge and is a significantly stronger nucleophile than neutral phenol [1].	2
20(a)	Alkaline potassium manganate(VII), $\text{KMnO}_4 / \text{OH}^-$ (or acidified $\text{KMnO}_4 / \text{H}^+$) [1]; Heated under reflux [1].	2
20(b)	Purple solution forms a dark brown precipitate of MnO_2 (in alkaline medium) or decolourises (in acid) [1]; Upon cooling and adding dilute acid, white crystals / precipitate of benzoic acid separate out [1].	2
21(a)	Fluoroethanoic acid is more acidic (has lower pK_a) [1].	1
21(b)	Fluorine is more electronegative than chlorine [1]; Fluorine exerts a greater electron-withdrawing inductive ($-I$) effect [1]; This withdraws electron density more effectively from the $-\text{COO}^-$ group, further stabilising the fluoroethanoate conjugate base [1].	3
22(a)	Add water at room temperature [1]; Ethanoyl chloride reacts violently with steamy acidic fumes and heat evolved; chloroethane is immiscible and shows no reaction [1].	2
22(b)	Add aqueous silver nitrate at room temperature [1]; Ethanoyl chloride forms an immediate dense white precipitate of AgCl ; chloroethane produces no precipitate at room temperature [1].	2
23(a)	$-\text{CO}-(\text{CH}_2)_4-\text{CO}-\text{NH}-(\text{CH}_2)_6-\text{NH}-$ [2].	2
23(b)	Diacyl chlorides react much faster and at room temperature without requiring heat or acid catalysts [1]; The reaction goes to completion and gives high yields rapidly (e.g. interfacial polymerisation / Nylon rope trick) [1].	2
24(a)	Neutral [1].	1
24(b)	In amides, the lone pair of electrons on the nitrogen atom is in a p-orbital adjacent to the carbonyl π -bond [1]; The nitrogen lone pair delocalises into the carbonyl π -system by resonance [1]; This delocalisation significantly reduces electron density on nitrogen, making it unavailable to accept a proton (H^+) [1].	3
25(a)	$\text{CH}_3\text{COOH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{COCl} + \text{POCl}_3 + \text{HCl}$ [2].	2
25(b)	Phosphoryl chloride, POCl_3 , is a liquid with a boiling point (106°C) close to many acyl chlorides, requiring careful fractional distillation [1]; In contrast, SOCl_2 produces SO_2 and HCl which are both gases that escape spontaneously [1].	2
26(a)	Deuterated ethanoic acid, CH_3COOD [1]; Deuterium chloride, DCl [1].	2
26(b)	The broad O-H stretch at $2500\text{--}3000 \text{ cm}^{-1}$ is shifted to a significantly lower wavenumber ($\sim 1900\text{--}2200 \text{ cm}^{-1}$) for the O-D stretch [1]; Deuterium has double the mass of hydrogen, increasing reduced mass μ and lowering vibrational frequency [1].	2
27(a)	Hexanedioic acid has stronger London dispersion forces due to its longer carbon chain, but ethanedioic acid packs more closely into a rigid crystal lattice with extensive hydrogen bonding, giving it a relatively high melting point [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
27(b)	Hexanedioic acid contains a non-polar hydrophobic chain of four $-(CH_2)-$ groups which disrupts water's hydrogen bond network [1]; Ethanedioic acid consists almost entirely of polar carboxyl groups that form extensive hydrogen bonds with water [1].	2
28(a)	$C_6H_5COOC_6H_4-4-CH_3$ (4-methylphenyl benzoate) [2].	2
28(b)	4-methylphenyl benzoate [1]; NaOH deprotonates 4-methylphenol to form the strongly nucleophilic phenoxide anion and absorbs the HCl byproduct [1].	2
29(a)	Deep blue solution turns to form a red-brown / brick-red precipitate [2].	2
29(b)	Copper(I) oxide, Cu_2O [1]; Ethanoic acid lacks the oxidisable aldehydic C-H group present in methanoic acid [1].	2
30(a)	Hydrolysis of ethanoyl chloride has a much lower activation energy (E_a) than chloroethane [2].	2
30(b)	In ethanoyl chloride, nucleophilic attack forms a stable tetrahedral intermediate without breaking the strong C-Cl σ -bond in the rate-determining step [1]; In chloroethane, substitution requires breaking the C-Cl bond in the rate-determining transition state, demanding significantly higher energy [1].	2
31(a)	Propanone (acetone) [1]; $(CH_3COO)_2Ca \rightarrow CH_3COCH_3 + CaCO_3$ [1].	2
31(b)	Ethanal (acetaldehyde) [1]; Cross-condensation between methanoate (providing H^-) and ethanoate (providing acetyl) yields ethanal and $CaCO_3$ [1].	2
32(a)	$CH_3-C(=O)-COOH$ with all bonds displayed [1].	1
32(b)	The carbonyl group ($C=O$) adjacent to the carboxyl group is strongly electron-withdrawing ($-I$ inductive effect) [1]; This withdraws electron density from the carboxylate group, spreading and delocalising the negative charge on the pyruvate anion [1]; This stabilises the conjugate base relative to propanoate, greatly increasing K_a and lowering pK_a [1].	3
33(a)	Phosphorus(V) chloride / PCl_5 [1]; Phosphorus(III) chloride / PCl_3 [1].	2
34(a)	Vigorous reaction / effervescence [1]; Steamy / misty white fumes of hydrogen chloride (HCl) gas evolved [1].	2
35(a)	Carbon dioxide, CO_2 [1]; Water, H_2O [1].	2
36(a)	Primary amide (or carboxamide / $-CONH_2$) [1]; CH_3CONH_2 [1].	2
37(a)	$pK_a = -\log_{10}(K_a)$ [1]; A lower (or more negative) pK_a corresponds to a higher K_a and therefore a stronger acid [1].	2
38(a)	$CH_3COCl + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + HCl$ [2].	2
39(a)	$C_6H_5COOC_6H_5$ with ester linkage $-COO-$ joining two benzene rings correctly [2].	2
40(a)	Aqueous sodium hydroxide, $NaOH(aq)$ (or $KOH(aq)$) [1]; Heated under reflux [1].	2
41(a)	Ethanoic acid < Chloroethanoic acid < Trichloroethanoic acid [1].	1
41(b)	Electronegative chlorine atoms exert an electron-withdrawing inductive ($-I$) effect [1]; This withdraws electron density away from the negative $-COO^-$ group, spreading and delocalising the charge over a larger volume [1]; Greater delocalisation confers thermodynamic stability on the conjugate base, shifting equilibrium right and increasing K_a [1].	3
41(c)	Chloroethanoic acid is stronger than bromoethanoic acid [1]; Chlorine is more electronegative than bromine, exerting a greater electron-withdrawing inductive effect and better stabilising the carboxylate anion [1].	2
42(a)	Ethanoyl chloride: violent reaction with steamy white fumes and heat evolved [1]; 1-chlorobutane: immiscible layer, no visible reaction at room temperature [1]; Chlorobenzene: immiscible layer, no reaction even on boiling [1].	3
42(b)	In ethanoyl chloride, the carbonyl carbon has two electronegative atoms (O and Cl) creating a large δ^+ charge; addition of water is rapid and chloride is an excellent leaving group [1]; In chlorobenzene, a chlorine lone pair delocalises into the benzene π -ring, giving the C-Cl bond partial double bond character and making it very strong [1]; Furthermore, the incoming nucleophile is repelled by the high electron density of the benzene ring [1].	3
43(a)	Manganese changes from +7 in MnO_4^- to +2 in Mn^{2+} [1]; Carbon changes from +3 in $(COOH)_2$ to +4 in CO_2 [1].	2
43(b)	Moles of $MnO_4^- = 0.01820 \times 0.0200 = 3.64 \times 10^{-4} \text{ mol}$ [1]; Moles of $HCOOH = (5/2) \times 3.64 \times 10^{-4} = 9.10 \times 10^{-4} \text{ mol}$ [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
43(c)	Ethanoic acid does not contain a C-H bond on the carbonyl carbon (no aldehydic hydrogen) [1]; The C-C bond in ethanoic acid is stable and resistant to cleavage by KMnO_4 [1].	2
44(a)	Steam and concentrated H_3PO_4 catalyst at 300 °C, 60 atm (or HBr followed by warm NaOH(aq) for anti-Markovnikov route via hydroboration) [2].	2
44(b)	Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ heated under reflux to form propanoic acid [1]; Followed by SOCl_2 (or PCl_5) to form propanoyl chloride [1].	2
44(c)	Ethylamine, $\text{CH}_3\text{CH}_2\text{NH}_2$ [1]; $\text{CH}_3\text{CH}_2\text{COCl} + \text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3 + \text{HCl}$ [1].	2
45(a)	$\text{CH}_3\text{CH}_2\text{COCl} + 2\text{CH}_3\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_3 + \text{CH}_3\text{NH}_3^+ \text{Cl}^-$ (or $+ \text{CH}_3\text{NH}_2$ amide + HCl) [2].	2
45(b)	Nucleophilic addition-elimination (condensation) [1]; Methylamine possesses a lone pair of electrons on the nitrogen atom that attacks the electron-deficient carbonyl carbon [1].	2
46(a)	Mixture of sodium hydroxide (NaOH) and calcium oxide (CaO) [1]; Acts as a solid base that does not melt glass and decarboxylates the carboxylate group [1].	2
46(b)	$\text{CH}_3\text{COONa} + \text{NaOH} \rightarrow \text{CH}_4 + \text{Na}_2\text{CO}_3$ [1]; Methane, CH_4 [1].	2
47(a)	$-\text{O}-(\text{CH}_2)_3-\text{O}-\text{CO}-(\text{CH}_2)_2-\text{CO}-$ with correct ester linkages [2].	2
47(b)	Water, H_2O [1]; Condensation polymerisation [1].	1
48(a)	Add aqueous silver nitrate, $\text{AgNO}_3(\text{aq})$, directly (or add water then AgNO_3) [1]; Ethanoyl chloride produces an immediate white precipitate of AgCl ; ethanoic acid produces no precipitate [1].	2
48(b)	The ethanoyl chloride solution has a much lower pH (~1) than the ethanoic acid solution (~3) [1]; Ethanoyl chloride hydrolyses to produce HCl , a strong acid that ionises completely, whereas ethanoic acid is a weak acid that is only partially ionised [1].	2
49(a)	Sulfur dioxide, SO_2 [1]; Hydrogen chloride, HCl [1].	2
50(a)	A hydrogen atom attached directly to the carbonyl carbon atom (aldehydic hydrogen / $\text{H}-\text{C}=\text{O}$) (or adjacent $\text{C}=\text{O}$ groups as in ethanedioic acid) [2].	2